

Sistemas Multiagentes (PPGCA-CT e PPGCC-PG)

Artificial Intelligence: Introduction

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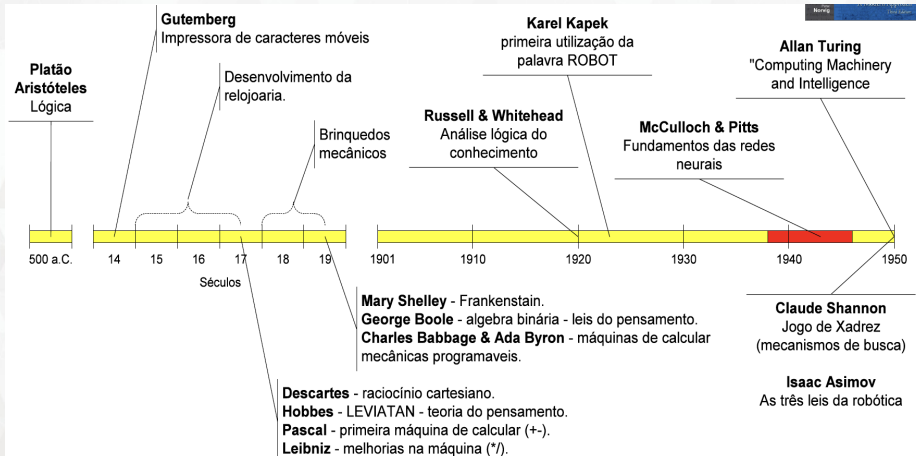
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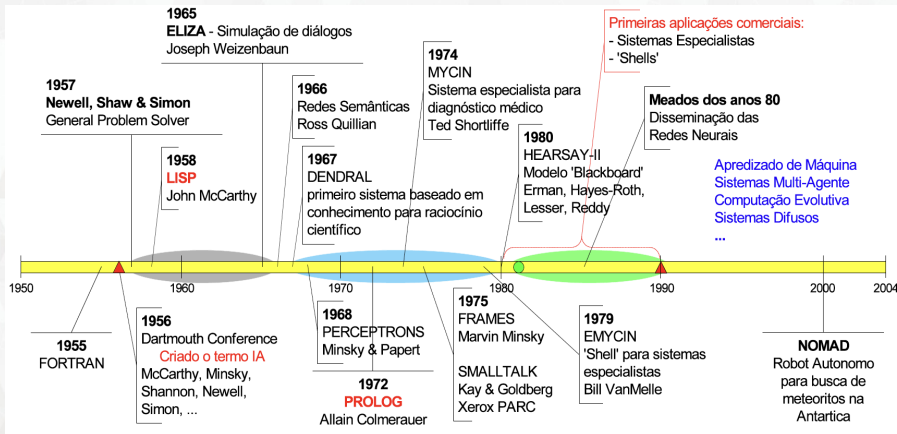
Agenda

- ① S. Russel & P. Norvig. AI: A Modern Approach, 4th ed., 2020 - Chap.1
 - ▶ The History
 - ▶ What is AI?
 - ▶ Foundations
- ② Mohamad Abou Ali, Fadi Dornaika, Jinan Charafeddine. Agentic AI: A comprehensive survey of architectures, applications, and future directions. Artificial Intelligence Review, 59:11, 2026. doi: 10.1007/s10462-025-11422-4.
 - ▶ Historical Evolution of Five AI Paradigms
 - ① Symbolic AI
 - ② Machine Learning
 - ③ Deep Learning
 - ④ Generative AI
 - ⑤ Agentic AI
 - ▶ Two Lineages of Agentic AI & Key Message
- ③ Risks and Benefits
- ④ Major Global AI Efforts to Human Society

The History



The History



AI History Summary

- 1 The inception of AI (1943–1956)
 - ▶ Marvin Minsky (1969), John McCarthy (1971) defined the foundations of AI based on representation and reasoning
- 2 Early enthusiasm, great expectations (1952–1969)
 - ▶ LISP (John McCarthy, 1958), blocks world, perceptron convergence theorem (Block et al., 1962)
- 3 A dose of reality (1966–1973)
 - ▶ machine evolution (genetic programming)
- 4 Expert systems (1969–1986)
 - ▶ weak methods (general) do not scale up to large or difficult problem instances
 - ▶ Minsky's idea of frames (1975)
- 5 The return of neural networks (1986–present)
 - ▶ back-propagation learning algorithm of early 1960s
- 6 Probabilistic reasoning and machine learning (1987–present)
 - ▶ Hidden Markov models (HMMs) to speech recognition, Bayesian networks to represent uncertain knowledge, practical algorithms for probabilistic reasoning
- 7 Big data (2001–present)
 - ▶ very large data sets - trillions of words of text, billions of images, billions of hours of speech and video, vast amount of genomic data, vehicle tracking data, social network data, clickstream data, etc...
- 8 Deep learning (2011–present)
 - ▶ multilayer neural networks (Yoshua Bengio, Geoffrey Hinton, Yann LeCun, 2019)
 - ▶ convolutional neural networks - speech and visual object recognition

What is AI?

Historically, researchers have pursued different versions of AI.

- some have defined intelligence in terms of fidelity to human performance
- others prefer formal definition of intelligence called rationality — doing the “right thing”
- Rationality itself also varies
 - ▶ some consider intelligence to be a property of internal thought processes and reasoning
 - ▶ others focus on intelligent behavior, an external characterization¹

¹ In the public eye, there is confusion between the terms “AI” and “machine learning”

What is AI?

Thinking Humanly “The exciting new effort to make computers think ... <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI ... is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

2 dimensions: humanly vs. rationally & thinking vs. acting
4 possible combinations

Thinking Humanly: The cognitive modeling approach



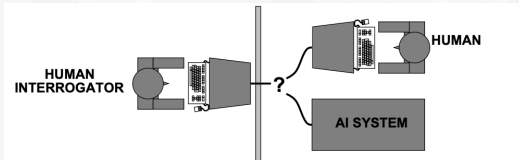
To say that a program thinks like a human, we must know how humans think, learning about human thought in 3 ways:

- introspection — trying to catch our own thoughts as they go by
- psychological experiments — observing a person in action
- brain imaging — observing the brain in action

The field of **Cognitive Science** brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind

Acting Humanly: The Turing test

- Alan Turing's article (1950)² - "Can a machine think?"
- The Imitation Game - operational test for intelligent behavior



- Predicted by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Suggested major components of AI:
 - ▶ natural language processing - communicate in human language
 - ▶ knowledge representation - store what it knows or hears
 - ▶ automated reasoning - answer questions & draw new conclusions
 - ▶ machine learning - adapt, detect, extrapolate patterns
- Other researchers propose total Turing Test
 - ▶ computer vision & speech recognition - perceive the world
 - ▶ robotics - manipulate objects & move about

Thinking Rationally: The “laws of thought” approach



- Aristotle (Greek philosopher) attempt to codify “right thinking” — reasoning processes - correct conclusions when given correct premises
- Canonical example - Socrates is a man and all men are mortal then Socrates is mortal
- These laws of thought were supposed to govern the operation of the mind - their study initiated the field called logic
- Direct line through mathematics and philosophy to modern AI
- Problem: Not all intelligent behavior is mediated by logical deliberation + complete knowledge of the world is seldom achieved (probability theory with uncertain information)

Acting Rationally: The rational agent approach

- Agent acts (Latin *agere*, to do) doing the right thing
- The right thing: which is expected to maximize goal achievement given the available information, when there is uncertainty, the best expected outcome
- Not necessarily involve thinking (e.g., blinking reflex) but thinking should be in the service of rational action
- Perfect rationality often remains a good starting point for theoretical analysis

Rational Agents

- This course is about designing rational agents
- Agent software entity that perceives and acts
- Abstractly, function from percept histories to actions

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$

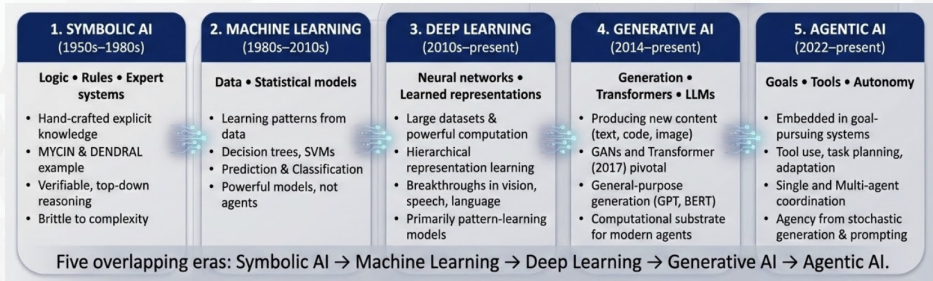
- For any given class of environments & tasks, we seek the agent (or class of agents) with the best performance
- Computational limitations make perfect rationality unachievable
- Design best program for given machine resources

AI Foundations

- 1 **Philosophy**
 - ▶ dualism, empiricism, induction, logical positivism
 - ▶ confirmation theory, utility, utilitarianism
 - ▶ deontological ethics
- 2 **Mathematics**
 - ▶ formal logic
 - ▶ algorithm, computability, incompleteness, tractability, NP-completeness
 - ▶ probability, statistics
- 3 **Economics**
 - ▶ decision theory (game theory, multiagent systems), operations research (Markov decision processes), satisficing
- 4 **Neuroscience**
 - ▶ neuron, optogenetics, brain-machine interface, singularity
- 5 **Psychology**
 - ▶ behaviorism, cognitive psychology, cognitive science
 - ▶ human-computer interaction, intelligence augmentation
- 6 **Computer engineering**
 - ▶ Moore's law, quantum computing
- 7 **Control theory and cybernetics**
 - ▶ homeostatic, cost function
- 8 **Linguistics**
 - ▶ knowledge representation
 - ▶ computational linguistics, natural language processing

Historical Evolution of AI Paradigms³

- a systematic PRISMA-based review of 90 studies (2018–2025)
- provides a conceptual toolkit to guide future research, development & policy toward robust, trustworthy hybrid intelligent systems



The transition is driven by changes in how intelligence is **represented**, **learned**, **generated**, and ultimately **enacted**.

³ M. A. Ali, F. Dornaika, J. Charafeddine. Agentic AI: A comprehensive survey of architectures, applications, and future directions. AI Review, 59:11, 2026

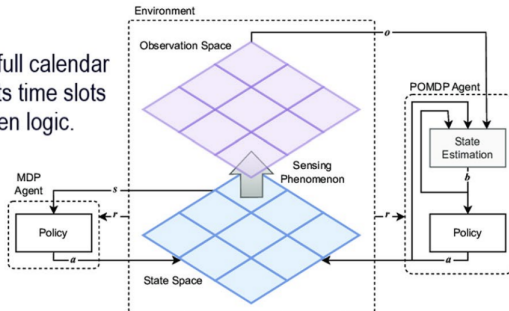
Symbolic AI (1950s–1980s)

- Intelligence - represented through logic, explicit knowledge, hand-crafted rules.
- Expert systems - MYCIN & DENDRAL exemplified top-down, deductive reasoning.
- Later symbolic architectures - MDPs, POMDPs, BDI, and SOAR⁴.
- Strengths - explicit reasoning, planning, persistent state, and verifiability.
- Limitation - brittleness and difficulty scaling to complex, changing environments.

⁴ The SOAR cognitive architecture by John E. Laird (2022) An analysis and comparison of Adaptive Control of Thought–Rational (ACT-R) and SOAR. arXiv:2201.09305. Presented 9th Annual Conference Advances in Cognitive Systems (ACS'21).

From Rule-Based Scheduling to Belief-Based Inference

Robot sees full calendar
and suggests time slots
with if-then logic.



Assistant learns
hidden user
preferences from
declines and
acceptances.

Fig. 3 Classical symbolic reasoning: comparison between a rule-based MDP scheduler (left) and a belief-based POMDP assistant (right). The MDP agent relies on explicit calendar states and deterministic policies, while the POMDP agent infers hidden user preferences from behavioral feedback. Both represent the symbolic paradigm's approach to decision-making

Machine Learning - Deep Learning (1980s–2010s)

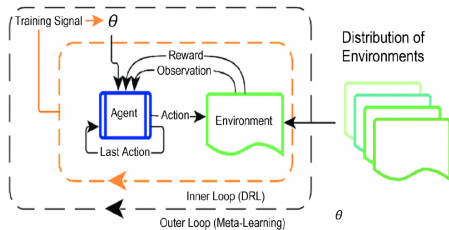
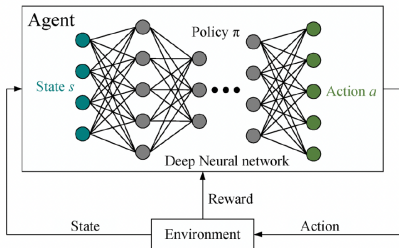


- Machine Learning (1980s–2010s) - shifted from hard-coded logic toward learning from data.
- Decision trees, Support Vector Machines (SVM), other statistical models enabled prediction, classification, and recommendation.
- Deep Learning (2010s–present) - combined large datasets and increased computational power with deep neural networks.
- Convolutional & recurrent architectures - enabled automated hierarchical representation learning in vision, speech, and language.
- Despite major advances - these systems were primarily powerful pattern-learning models rather than autonomous agents.

Machine Learning - Deep Learning (1980s–2010s)



- DRL is a bridge using neural networks to learn policies that symbolic systems explicitly program.
- Meta-DRL introduced generalization across tasks, a precursor to adaptability required for modern agency.



Generative AI (2014–present)

- Generative AI - moved beyond perception & classification toward producing new content.
- Generative Adversarial Networks (GANs) - an early milestone & Transformer architecture (2017) became pivotal for scaling Large Language Models (LLMs).
- LLMs - GPT & BERT enabled general-purpose generation of text, code, and other media.
- Generative models - provided the computational substrate for modern Agentic AI.

From Basic Scheduling to Intuitive Personal Assistant

- The evolution of a personal assistant from a deterministic rule-based system.
- To an uncertainty-aware (POMDP) system.
- Finally, a modern LLM orchestrated agent.

MDP Phase:

The Robot Scheduling



POMDP Phase:

The Mind-Reading Assistant



Infers Hidden Preferences
Deals with Uncertainty



Meta-Learning:

The Super-Adaptive AI Companion



Fast Adaptation

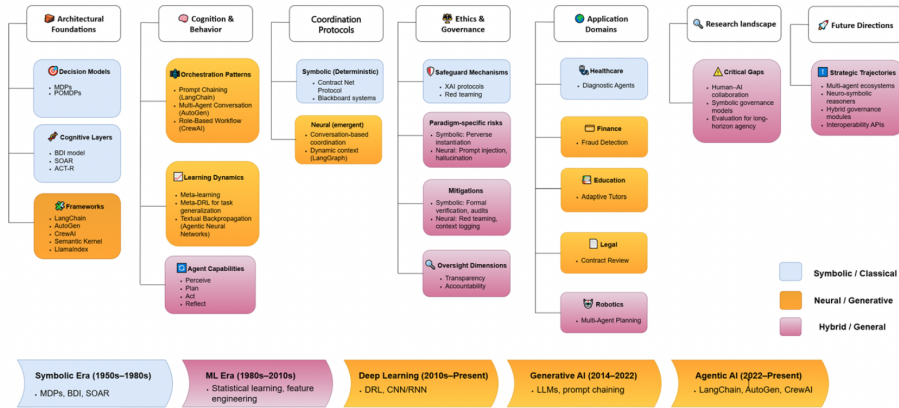
Learns How to Learn

Adapts Quickly to
New Users

Agentic AI (2022–present)

- Generative models - embedded in systems that pursue goals, use tools, plan tasks, and adapt through environmental feedback.
- Single agents - decompose & execute multi-step tasks, MAS coordinate specialized agents.
- Popular frameworks - AutoGPT, AutoGen, CrewAI, LangGraph.
- Agency in the neural lineage emerges through stochastic generation and prompt-driven orchestration rather than necessarily through symbolic planning.

Paradigm-annotated Taxonomy of Agentic AI Systems



7 modular domains, color-coded by architectural paradigm: symbolic/classical (blue), neural/generative (orange), and hybrid/general (purple).

Two Lineages of Agentic AI⁵

SYMBOLIC / CLASSICAL

- Algorithmic planning
- Explicit state and rules
- Persistent symbolic representations
- BDI, SOAR, POMDPs
- Reliability and verifiability

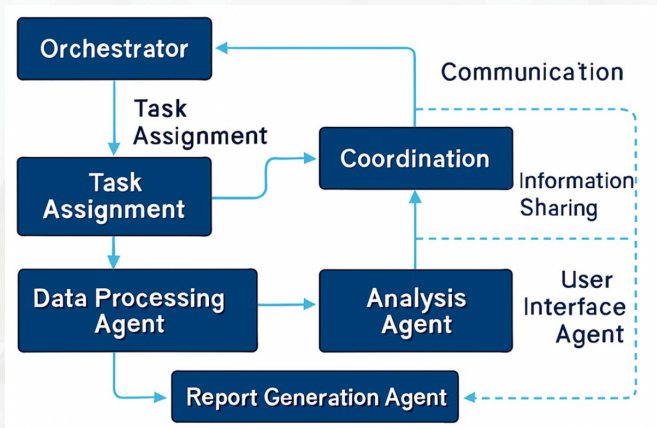
NEURAL / GENERATIVE

- Stochastic generation
- Prompt-driven orchestration
- LLM context and tool use
- AutoGen, CrewAI, LangGraph
- Adaptability and generalization

The survey warns against “conceptual retrofitting”: applying symbolic architectural concepts to systems built on fundamentally different neural mechanisms.

⁵ M. A. Ali, F. Dornaika, J. Charafeddine. Agentic AI: A comprehensive survey of architectures, applications, and future directions. *AI Review*, 59:11, 2026

Multi-agent Orchestration in Modern AI Systems



Most advanced manifestation of neural paradigm is multi-agent orchestration, that coordinate diverse, modular agents through structured communication protocols.

Key Message⁶

- AI evolution is not a simple replacement of paradigms.
- Different paradigms remain appropriate $\neq$ requirements $\leftrightarrow$ paradigm choice is strategic.
- Symbolic AI emphasizes explicit reasoning, planning, state, and verifiability.
 - ▶ symbolic systems dominate safety-critical domains (e.g., healthcare).
- Neural AI emphasizes learning, representation, generation, and adaptability.
 - ▶ neural systems prevail in adaptive, data-rich environments (e.g., finance).
- Agentic AI adds goal-directed autonomy, tool use, memory/context, and coordinated action.
- The survey identifies critical research gaps:
 - ▶ a significant deficit in governance models for symbolic systems.
 - ▶ a pressing need for hybrid neuro-symbolic architectures, important direction for combining adaptability with reliability.

⁶ M. A. Ali, F. Dornaika, J. Charafeddine. Agentic AI: A comprehensive survey of architectures, applications, and future directions. AI Review, 59:11, 2026

Risks and Benefits

- AI plays an increasingly important role in the economic, social, scientific, medical, financial, and military spheres, but what are risks & benefits?
- Risks from the misuse of AI
 - ▶ Lethal autonomous weapons - weapons that can locate, select, and eliminate human targets without human intervention
 - ▶ Surveillance and persuasion - concern that became apparent in elections beginning in 2016
 - ▶ Biased decision making - race, gender, or other protected categories
 - ▶ Impact on employment - machines eliminating jobs
 - ▶ Safety-critical applications - driving cars, water supplies managing in cities
 - ▶ Cybersecurity - AI techniques contribute to the potency, survivability, and proliferation capability of malware
 - ▶ Over the past few years, researchers have publicly resigned from AI labs, warning that they are racing headfirst towards catastrophe.
 - ▶ On Sept 8th, 2026 - Jacob Coxon, British Softw Engineer, left Anthropic (OpenAI competitor) as the enterprise is moving towards systems that can surpass human control capabilities!

Major Global AI Efforts to Human Society

- 1 Stanford University has 100 year study on AI (AI100) convenes panels of experts to provide reports to help track progress.
- 2 2026 Artificial Intelligence Index Report produced by Stanford Institute for Human-Centered Artificial Intelligence (HAI).⁷
 - ▶ Society is at a crucial juncture in determining how to deploy AI-based technologies in ways that promote rather than hinder democratic values such as freedom, equality, and transparency
- 3 2024-2028 MCTI lança na 5a Conferência Nacional CT&I -Plano Brasileiro de Inteligência Artificial (PBIA) - IA para o bem de todos, prevê investimentos de R\$ 23 bilhões para tornar o país referência em inovação, soberania digital e inclusão social.⁸
- 4 2024 A Sociedade Brasileira de Computação (SBC) lançou o Plano de IA, que traz um diagnóstico do setor e diretrizes estratégicas para o uso da tecnologia no Brasil.⁹

⁷ https://hai.stanford.edu/assets/files/ai_index_report_2026.pdf

⁸ <https://pbia.cgee.org.br>

⁹ <https://doi.org/10.5753/sbc.rt.2024.141>

Thanks

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